

# Tool Selection — Build vs Buy

Generative AI for Business Leaders & C-Suite | AI Implementation Strategy

## 1. The Default Answer Is Buy

For business leaders evaluating AI adoption, the build-versus-buy decision is one of the most consequential strategic choices in the first year. The data is unambiguous: for ninety percent of use cases, buying existing AI solutions dramatically outperforms building custom ones on every dimension that matters — speed, cost, risk, and time to value.

The instinct to build is understandable. Leaders want control, differentiation, and intellectual property. Engineers want interesting problems. But these desires collide with a harsh reality: seventy to eighty percent of custom AI projects fail to reach production. They run over budget, miss timelines, or simply do not deliver expected results. Meanwhile, commercial AI platforms deliver working capabilities in days, not months.

**Key Point:** Default to buying AI tools unless you have exceptional, clearly articulated reasons to build. The burden of proof is on building, not buying. This single decision can save your organisation millions of dollars and years of wasted effort.

## 2. The Economics of Build vs Buy

The cost differential between building and buying AI is not marginal — it is often one to two orders of magnitude.

| Dimension              | Buy (Commercial Tools)                                    | Build (Custom AI)                                                  |
|------------------------|-----------------------------------------------------------|--------------------------------------------------------------------|
| Upfront cost           | \$20–\$60 per user/month                                  | \$500,000–\$5,000,000+ in development                              |
| Time to first value    | Days to weeks                                             | 6–18 months                                                        |
| Risk of failure        | Low — cancel subscription if tool underperforms           | 70–80% of projects fail to reach production                        |
| Ongoing maintenance    | Included in subscription                                  | \$100,000–\$500,000+/year for engineering, infrastructure, updates |
| Improvement trajectory | Vendor invests billions in R&D; you benefit automatically | Every improvement requires your own engineering resources          |
| Talent requirements    | Training existing staff on tool usage                     | Hiring scarce, expensive specialists (\$150K–\$400K+ per engineer) |
| Scalability            | Vendor handles infrastructure scaling                     | You manage servers, load balancing, failover, and monitoring       |

For a typical mid-size company with one hundred employees, comprehensive AI tool coverage costs thirty thousand to one hundred thousand dollars annually. That is less than one mid-level employee's

fully loaded cost. Even at the high end of commercial tool spending, you are investing a fraction of what a single custom AI project would consume.

### 3. Four Categories of AI Tools Worth Buying

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Commercial AI tools fall into four categories, each addressing different organisational needs. Understanding this taxonomy helps leaders make targeted purchasing decisions rather than defaulting to a single general-purpose platform.

#### ***Category 1 — General AI Assistants***

These handle writing, analysis, research, summarisation, and general knowledge work across all departments. They are the broadest-impact tools and should be your first purchase.

| Tool                         | Cost (per user/month) | Best For                                                                                  |
|------------------------------|-----------------------|-------------------------------------------------------------------------------------------|
| ChatGPT Enterprise           | \$60                  | Writing, analysis, code generation, research; strong reasoning and creative capability    |
| Microsoft Copilot            | \$30                  | Organisations already on Microsoft 365; deep integration with Word, Excel, Outlook, Teams |
| Google Workspace AI (Gemini) | \$20–\$30             | Google Workspace shops; embedded in Docs, Sheets, Gmail, Meet                             |

#### ***Category 2 — Specialised Functional AI***

Pre-built AI solutions designed for specific business functions, with domain-specific workflows and integrations already in place.

| Function         | Example Tools        | Cost Range             |
|------------------|----------------------|------------------------|
| Customer Service | Intercom, Zendesk AI | \$50–\$150/user/month  |
| Sales Enablement | Gong, Salesloft      | \$100–\$200/user/month |
| HR & Recruiting  | HireVue, Eightfold   | \$50–\$150/user/month  |
| Marketing        | Jasper, Copy.ai      | \$40–\$125/user/month  |

#### ***Category 3 — Industry-Specific AI Platforms***

These come with domain expertise and regulatory compliance built in, which is critical for heavily regulated industries.

- **Veeva:** Pharmaceutical and life sciences — manages clinical trials, regulatory submissions, and commercial operations with industry-specific AI.
- **nCino:** Banking and financial services — AI-powered loan origination, risk assessment, and compliance management.
- **Procore:** Construction — project management, safety compliance, and resource optimisation with built-in industry standards.

#### ***Category 4 — AI Features in Existing Enterprise Software***

Many enterprise platforms you already use are embedding AI capabilities. These require no new vendor relationships and often no additional licensing.

- **Salesforce Einstein:** Predictive lead scoring, automated opportunity insights, and natural-language querying of CRM data.
- **SAP AI:** Intelligent automation across procurement, supply chain, and financial operations.
- **Oracle AI:** Embedded analytics and automation across ERP, HCM, and cloud infrastructure.

## 4. The Three Scenarios Where Building Makes Sense

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Building custom AI is justified in exactly three scenarios. Leaders should be sceptical of anyone — including internal engineers — who claims additional scenarios exist.

### *Scenario 1 — Truly Unique Proprietary Data*

Your organisation possesses data that provides significant competitive advantage, and exposing it to third-party AI vendors creates unacceptable business or security risk. The critical test: is the data itself the competitive advantage, or is it what you do with the data? If it is the latter, commercial tools with strong data-privacy guarantees (such as ChatGPT Enterprise's data isolation) may still work.

Examples: a hedge fund's proprietary trading signals, a pharmaceutical company's unpublished drug-interaction research, or a defence contractor's classified operational data.

### *Scenario 2 — No Existing Tool Addresses the Use Case*

You have thoroughly researched available options and literally no commercial tool addresses your specific need. This is rarer than most leaders believe. Most companies overestimate how unique their needs are. Before concluding that no tool exists, require your team to evaluate at least ten commercial alternatives and document why each falls short.

### *Scenario 3 — AI Is Core to Your Revenue Product*

If you are selling AI capabilities to customers — not just using AI internally for operational efficiency — you obviously need to build or deeply customise. A company whose product is an AI-powered medical diagnostic platform cannot simply resell ChatGPT. But if AI is helping your internal teams work faster, buying is almost always right.

**Key Point:** Even in these three scenarios, the smartest approach is often hybrid. Start by buying general AI tools to build organisational capability. Then identify the one or two specific areas where custom AI creates defensible competitive advantage. Use the ROI from bought tools to fund selective custom development.

## 5. The Hybrid Model — Best of Both Worlds

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The most successful AI strategies are neither pure-build nor pure-buy. They are hybrids that use commercial tools for ninety percent of use cases and invest custom development resources only where they create disproportionate competitive advantage.

The hybrid progression looks like this:

- **Phase 1 (Months 1–3):** Buy general-purpose AI tools for broad access. Build organisational AI literacy and identify high-value use cases through experimentation.
- **Phase 2 (Months 4–9):** Add specialised bought tools for validated departmental use cases. Begin collecting proprietary data that might fuel custom AI later.
- **Phase 3 (Months 10–18):** Evaluate whether any use case requires custom AI. If yes, fund development from proven AI savings. If no, continue scaling commercial tools.

## 6. The First Ninety Days — Your Tool Strategy

A practical, week-by-week tool strategy for the first ninety days eliminates analysis paralysis and gets AI into employees' hands quickly.

| Timeline   | Action                                                                                                              | Budget                             |
|------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Week 1     | Select and deploy one general AI assistant (ChatGPT Enterprise or Microsoft Copilot) for all pilot participants     | \$1,000–\$3,000/month              |
| Week 2     | Identify specialised tools needed for specific pilot projects (e.g., Intercom for customer service, Gong for sales) | Free trials or \$500–\$2,000/month |
| Weeks 3–12 | Use tools intensively, measure results rigorously, make keep-or-kill decisions based on data                        | Varies by tool adoption            |
| Day 90     | Decision point: which tools deliver ROI and deserve continued investment? Which should be cancelled?                | Total spend: \$10,000–\$30,000     |

Compare this \$10,000–\$30,000 ninety-day spend with the \$500,000–\$5,000,000 cost of custom development that would not even be finished yet. The buy-first approach does not preclude building later — it ensures you build the right things, informed by real usage data.

## 7. Red Flags That You Are Building When You Should Buy

Several warning signs indicate an organisation is defaulting to custom development for the wrong reasons:

- **"Our needs are unique":** The most dangerous phrase in AI strategy. Most business processes (sales, marketing, customer service, HR, finance) are not unique enough to require custom AI. Challenge this claim with evidence.
- **"We need full control over the technology":** Control is a valid concern, but enterprise AI vendors (ChatGPT Enterprise, Azure OpenAI) offer data isolation, SOC 2 compliance, and no-training-on-your-data guarantees that address most control requirements.
- **"Our engineers want to build it":** Engineer preference is not a business justification. Engineers naturally prefer building to buying because building is more interesting work. But their job is to create business value, not to satisfy intellectual curiosity.
- **"We do not trust third-party AI vendors":** If this is a genuine security concern, evaluate vendor security certifications rigorously. If it is a reflexive objection, it is costing you competitive advantage.

**Real-World Incident — Uber's Self-Driving Division (2016–2020):** Uber spent over \$2.5 billion building custom self-driving AI from 2016 to 2020. Despite massive investment in specialists and infrastructure, the

technology did not reach commercial viability. In 2020, Uber sold the entire division to Aurora Innovation at a significant loss. The lesson for business leaders: even well-funded custom AI efforts can fail spectacularly. Uber would have been better served by partnering with existing autonomous-vehicle technology providers while focusing on its core ride-hailing platform.

## 8. Vendor Evaluation Framework

When buying AI tools, use a structured evaluation to avoid impulsive purchases and ensure tools meet enterprise requirements.

| Criterion               | What to Evaluate                                                                                                              | Weight   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------|
| Data Security & Privacy | SOC 2 certification, data isolation, no-training-on-your-data policy, GDPR/CCPA compliance, encryption at rest and in transit | Critical |
| Integration             | API availability, existing integrations with your tech stack (CRM, ERP, HRIS), SSO/SAML support                               | High     |
| Total Cost of Ownership | Per-user licensing, implementation costs, training costs, API usage fees, contract terms                                      | High     |
| Vendor Viability        | Revenue trajectory, funding, customer base, product roadmap, market position                                                  | Medium   |
| User Experience         | Ease of adoption, learning curve, mobile accessibility, customisation options                                                 | Medium   |
| Support & SLA           | Response times, dedicated account management, uptime guarantees, escalation paths                                             | Medium   |

## 9. Key Terms Glossary

| Term                                    | Definition                                                                                                                                                                          |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Build vs Buy                            | The strategic decision between developing custom AI solutions in-house versus purchasing commercial off-the-shelf platforms                                                         |
| Total Cost of Ownership (TCO)           | The complete cost of an AI solution including licensing, implementation, training, maintenance, and opportunity cost of delayed deployment                                          |
| Proof of Concept (PoC)                  | A small-scale test of an AI tool or approach designed to validate feasibility and business value before committing to full deployment                                               |
| Data Isolation                          | A vendor guarantee that customer data is stored and processed separately from other customers' data and is not used to train the vendor's AI models                                 |
| SOC 2 Compliance                        | A widely recognised security certification demonstrating that a vendor meets rigorous standards for data security, availability, processing integrity, confidentiality, and privacy |
| API (Application Programming Interface) | A set of protocols allowing different software systems to communicate; critical for integrating AI tools with existing enterprise systems                                           |

|                    |                                                                                                                                                                     |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hybrid AI Strategy | An approach that combines commercial AI tools for most use cases with selective custom development only where it creates defensible competitive advantage           |
| Vendor Lock-in     | The risk that switching from one AI vendor to another becomes prohibitively expensive or disruptive due to data migration, workflow dependencies, or contract terms |

## 10. Quick Revision Checklist

- For 90% of use cases, buying AI tools dramatically outperforms building custom solutions on speed, cost, and risk
- 70–80% of custom AI projects fail to reach production
- A mid-size company can achieve comprehensive AI tool coverage for \$30,000–\$100,000/year — less than one employee's salary
- Four categories of buyable AI: general assistants, specialised functional tools, industry-specific platforms, and embedded features in existing enterprise software
- Only three scenarios justify building custom AI: truly unique proprietary data, no existing tool addresses the use case, or AI is core to your revenue product
- The hybrid model is the smartest approach: buy for 90%, build selectively for the 10% that creates competitive advantage
- First ninety days tool strategy: deploy a general assistant in week one, add specialised tools in week two, measure intensively for ten weeks, decide at day ninety
- Total ninety-day tool spend is typically \$10,000–\$30,000 versus \$500,000–\$5,000,000 for custom development
- Red flags for unnecessary building: 'our needs are unique', 'our engineers want to build it', 'we need full control'
- Use a structured vendor evaluation covering security, integration, TCO, viability, UX, and support

## 20 Exam-Based MCQs — Build vs Buy

*Test yourself after reading. These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.*

**Q1. What percentage of custom AI projects fail to reach production?**

- A) 20–30%
- B) 40–50%
- C) 70–80%
- D) 90–95%

**Answer: C** **Explanation:** C is correct — industry data shows 70–80% of custom AI projects fail to reach production due to budget overruns, missed timelines, or failure to deliver expected results. A and B significantly underestimate the failure rate. D overstates it.

**Q2. Scenario:** A logistics company's CTO proposes spending \$3 million to build a custom AI-powered route optimisation system. The COO notes that three commercial route-optimisation platforms exist at \$50–\$100 per vehicle per month. The fleet has 200 vehicles. What is the STRONGEST argument for the COO's position?

- A) Commercial tools are always better than custom solutions for logistics
- B) At \$100/vehicle/month for 200 vehicles, the annual cost is \$240,000 — the company could buy twelve years of commercial service for the price of one custom build with its 70–80% failure risk
- C) The CTO does not have the technical expertise to oversee a custom build
- D) Building custom AI would require hiring drivers with AI skills

**Answer: B** **Explanation:** B is correct because it quantifies the cost comparison and highlights the risk differential. \$240,000/year for proven, working software versus \$3 million with a 70–80% chance of failure is a clear business case. A overgeneralises. C attacks the person, not the argument. D is irrelevant — drivers do not need AI skills.

**Q3. Which of the following is a valid reason to build custom AI rather than buying a commercial solution?**

- A) Your engineering team is enthusiastic about AI and wants challenging projects
- B) You believe your business processes are unique (without evidence)
- C) You have proprietary data that provides competitive advantage and cannot be exposed to third-party vendors
- D) Your competitor recently announced a custom AI initiative

**Answer: C** **Explanation:** C is correct — truly unique proprietary data that creates competitive advantage and cannot be safely shared with vendors is one of the three valid scenarios for building. A is engineer preference, not a business justification. B is the 'our needs are unique' red flag without supporting evidence. D is a reactive, fear-driven response.

**Q4. Scenario:** A hospital network's CISO rejects all commercial AI tools, stating that patient data cannot be processed by any third-party system. The CIO points out that several healthcare-specific AI vendors hold HIPAA Business Associate Agreements, SOC 2 certification, and provide data isolation guarantees. What is the BEST path forward?

- A) Accept the CISO's position and build all AI tools in-house
- B) Override the CISO and deploy commercial tools without security review
- C) Evaluate healthcare-specific AI vendors with appropriate compliance certifications, involving the CISO in the security review process
- D) Delay AI adoption until regulations provide clearer guidance on AI in healthcare

**Answer: C** **Explanation:** C is correct because it addresses the legitimate security concern through evidence-based vendor evaluation rather than blanket rejection or blanket override. A surrenders to an uninformed objection without evaluating compliant alternatives. B ignores legitimate security responsibilities. D uses regulatory uncertainty as an excuse for inaction — many healthcare organisations are already using compliant AI tools.

**Q5. What is the estimated annual cost for comprehensive AI tool coverage for a typical mid-size company with 100 employees?**

- A) \$5,000–\$10,000
- B) \$30,000–\$100,000
- C) \$250,000–\$500,000
- D) \$1,000,000+

**Answer: B** **Explanation:** B is correct — comprehensive coverage across general AI assistants and specialised functional tools costs \$30,000–\$100,000/year for 100 employees, which is less than one mid-level employee's fully loaded cost. A is too low for 100 users across multiple tools. C and D represent custom-build territory.

**Q6. *Scenario:*** A consumer packaged goods company deploys ChatGPT Enterprise for \$60/user/month to 50 employees across marketing, sales, and operations. After three months, the marketing team reports a 40% reduction in content creation time and the sales team reports a 25% improvement in proposal quality. The CFO asks whether the \$9,000/month spend is justified. How should the success metrics owner respond?

- A) Cancel the subscription because \$9,000/month is too expensive for an AI tool
- B) Present the documented productivity gains and calculate the equivalent labour-hour savings to demonstrate ROI
- C) Expand the subscription to all 500 employees immediately without further analysis
- D) Switch to a cheaper AI tool to reduce costs

**Answer: B** **Explanation:** B is correct because quantified ROI is the appropriate response to a cost-justification question. If 50 employees save even 5 hours/month each at an average loaded cost of \$50/hour, that is \$12,500/month in productivity gains against \$9,000/month in tool cost — a clear positive ROI. A ignores the demonstrated value. C skips the analysis step. D optimises for cost rather than value.

**Q7. What is the recommended first step in the ninety-day tool strategy?**

- A) Conduct a six-month evaluation of all available AI tools before selecting one
- B) Select and deploy one general AI assistant platform for all pilot participants in week one
- C) Build a custom proof-of-concept AI tool to test feasibility
- D) Hire an AI consulting firm to recommend the right tools

**Answer: B** **Explanation:** B is correct — the ninety-day strategy starts with deploying a general AI assistant (ChatGPT Enterprise or Microsoft Copilot) in week one to get AI into employees' hands immediately. A delays action with excessive evaluation. C starts with building rather than buying. D introduces external dependency and delays deployment.



**Q8. Scenario:** A bank's engineering director argues for building a custom AI fraud-detection system, estimating \$2 million in development costs over 12 months. When challenged, he says: 'Our fraud patterns are unique — no commercial tool can match our data.' The risk officer notes that the bank currently uses a commercial fraud-detection platform that catches 94% of fraudulent transactions. What question should leadership ask before approving the custom build?

- A) How many engineers will need to be hired to complete the project?
- B) Can the existing commercial platform be tuned or upgraded to close the remaining 6% gap, and what would that cost compared to \$2 million?
- C) What programming language will the custom system be built in?
- D) Has the engineering director built a similar system at a previous company?

**Answer: B** **Explanation:** B is correct because it tests the 'our needs are unique' claim against evidence. If the commercial platform can be tuned to improve from 94% to 97% for \$200,000, that is far more cost-effective than a \$2 million custom build with a 70–80% failure risk. A, C, and D are implementation details that should only matter after the strategic decision is made.

**Q9. Which category of AI tools should typically be purchased FIRST in an enterprise AI adoption programme?**

- A) Industry-specific AI platforms
- B) Specialised functional AI tools
- C) General AI assistants like ChatGPT Enterprise or Microsoft Copilot
- D) Custom-built AI solutions

**Answer: C** **Explanation:** C is correct because general AI assistants provide the broadest impact across all departments, build organisational AI literacy, and help identify which specialised tools to purchase next. A and B are narrower and should follow the general assistant. D is not a purchase category — it is a build decision.

**Q10. Scenario:** A retail chain's CEO reads about a competitor building custom AI for demand forecasting. She tells her team: 'We need to build our own AI too — we cannot fall behind.' The data analytics manager responds that their existing ERP system (SAP) already includes AI-powered demand forecasting that they have not activated yet. What does this scenario illustrate about the build-vs-buy decision?

- A) The CEO is correct — competitive pressure justifies custom AI investment
- B) Many organisations already have AI capabilities embedded in their existing enterprise software that they have not yet activated — buying may mean simply turning on features they already own
- C) ERP-embedded AI is always inferior to custom-built solutions
- D) The competitor's custom AI initiative will likely succeed and provide lasting advantage

**Answer: B** **Explanation:** B is correct — Category 4 AI tools (embedded features in existing enterprise software) are often overlooked. Activating SAP's built-in demand forecasting costs nothing incrementally and can be done immediately. A is a reactive decision driven by competitive fear rather than analysis. C overgeneralises. D assumes the competitor's custom build will succeed despite the 70–80% failure rate.

**Q11. What is the PRIMARY risk of 'vendor lock-in' when buying AI tools?**

- A) The vendor may increase prices without warning
- B) Switching to a different vendor becomes prohibitively expensive or disruptive due to data migration, workflow dependencies, and retraining
- C) The vendor may go bankrupt, leaving you without AI capabilities
- D) Employees may become too dependent on the vendor's specific interface

**Answer: B** **Explanation:** B is correct — vendor lock-in refers to the economic and operational difficulty of switching providers once workflows, data, and training are built around a specific platform. A is a pricing risk, not lock-in specifically. C is a viability risk. D describes habit formation, not the structural dependencies that define lock-in.

**Q12. *Scenario:*** A media company's product team wants to integrate AI-generated content into their consumer-facing app. The content will be a paid feature that customers subscribe to. The CTO suggests using OpenAI's API to power the feature. Does this scenario meet the criteria for building custom AI?

- A) No — they should always use commercial tools regardless of the use case
- B) Yes, partially — since AI is core to a revenue product (Scenario 3), some customisation is justified, though they can still use a commercial API (like OpenAI) as the underlying engine with custom application logic on top
- C) Yes — they must build everything from scratch, including their own large language model
- D) No — they should license an existing media company's AI solution instead

**Answer: B** **Explanation:** B is correct because Scenario 3 (AI is core to a revenue product) applies, but this does not require building a model from scratch. Using a commercial API as the engine with custom application logic on top is the hybrid approach that provides product differentiation without unnecessary infrastructure investment. A ignores the revenue-product scenario. C demands unnecessary investment. D may not exist.

**Q13. What is the MOST important criterion when evaluating AI vendors for enterprise use?**

- A) The vendor's brand recognition and market share
- B) Data security and privacy protections, including SOC 2 certification and data isolation policies
- C) The lowest per-user price available
- D) The number of AI features the vendor advertises

**Answer: B** **Explanation:** B is correct — data security is weighted as 'Critical' in the vendor evaluation framework because a security breach or data misuse can cause catastrophic reputational and legal damage that no productivity gain can offset. A measures popularity, not suitability. C optimises for cost over value and security. D measures marketing, not capability.

**Q14. *Scenario:*** An insurance company's innovation lab has spent 18 months and \$4 million building a custom AI claims-processing system. The system works in testing but processes claims 15% slower than

*the existing manual process. The lab director requests an additional \$2 million and 12 months to optimise performance. What should leadership do?*

- A) Approve the additional funding — the sunk cost justifies continued investment
- B) Evaluate whether commercial claims-processing AI tools can achieve the goal faster and cheaper, and consider discontinuing the custom build
- C) Double the engineering team to accelerate development
- D) Launch the system as-is, despite being slower than manual processing

**Answer: B** **Explanation:** B is correct because \$4 million already spent is a sunk cost that should not influence future decisions. The relevant question is whether the next \$2 million is better spent on continuing a struggling custom build or on proven commercial alternatives. A falls for the sunk cost fallacy. C increases spending on a project that may be fundamentally flawed. D delivers negative value.

**Q15. In the hybrid AI strategy, what is the recommended approach for Phase 1 (months 1–3)?**

- A) Hire data scientists and begin building custom AI models
- B) Buy general-purpose AI tools for broad access and build organisational AI literacy
- C) Conduct a comprehensive AI maturity assessment before deploying any tools
- D) Identify and build the one custom AI solution that will provide competitive advantage

**Answer: B** **Explanation:** B is correct — Phase 1 focuses on buying general-purpose tools, giving employees access, and building AI literacy through real usage. This creates the organisational foundation for informed decisions about specialised tools (Phase 2) and selective custom builds (Phase 3). A skips validation. C delays action. D attempts custom development before the organisation has AI experience.

**Q16. *Scenario:* A pharmaceutical company's R&D division has proprietary molecular screening data that could train a custom AI model for drug discovery. The data represents 15 years of research and is considered a core competitive asset. Exposing it to third-party AI vendors would create unacceptable IP risk. Which build-vs-buy scenario does this represent?**

- A) Scenario 1 — Truly unique proprietary data that provides competitive advantage
- B) Scenario 2 — No existing tool addresses the use case
- C) Scenario 3 — AI is core to the revenue product
- D) None — they should still buy commercial tools

**Answer: A** **Explanation:** A is correct — this is a textbook Scenario 1 case: proprietary data that is itself a competitive advantage, and exposing it to third-party vendors creates unacceptable risk. B is plausible but the primary justification is data sensitivity, not tool unavailability. C would apply if they were selling AI-powered drug discovery as a product. D ignores the legitimate proprietary data concern.

**Q17. Why is 'our engineers want to build it' considered a red flag in the build-vs-buy decision?**

- A) Because engineers are not qualified to make strategic business decisions
- B) Because engineer preference for interesting technical challenges is not a valid business justification for multi-million-dollar investment decisions

- C) Because engineers always overestimate the cost of custom development
- D) Because only executives should decide what technology the company uses

**Answer: B** **Explanation:** B is correct — engineers naturally prefer building to buying because building is more intellectually stimulating. This is a professional preference, not a business case. The decision should be driven by cost, risk, speed-to-value, and competitive advantage, not by what the engineering team finds most interesting. A and D overstate the issue. C is the opposite — engineers often underestimate costs.

**Q18. Scenario:** A technology startup has 30 employees and \$5 million in annual revenue. The founder wants to build a custom AI system for internal sales forecasting. A board member suggests buying Salesforce Einstein instead. What is the MOST compelling reason the board member's suggestion is better?

- A) Salesforce is a larger company and therefore produces better AI
- B) A 30-person startup should preserve capital and engineering resources for its core product, not internal tooling that commercial platforms already handle well
- C) Custom AI never works for startups
- D) Salesforce Einstein is free

**Answer: B** **Explanation:** B is correct because startups have limited capital and engineering bandwidth, both of which should be directed toward the core product that generates revenue. Building custom internal AI diverts scarce resources from what matters most. A is an irrelevant comparison. C overgeneralises. D is incorrect — Salesforce Einstein requires licensing.

**Q19. What is the key advantage of buying AI tools that continuously improve through vendor R&D investment?**

- A) You can negotiate lower prices as the tool improves
- B) Your AI capabilities improve automatically without requiring your own engineering resources or additional investment
- C) Vendors are legally required to share their improvement roadmaps
- D) Continuous improvement means you never need to train employees on new features

**Answer: B** **Explanation:** B is correct — when vendors like OpenAI, Microsoft, or Google invest billions in R&D, their improvements flow to all customers automatically. If you built custom AI, every improvement would require your own engineering time and budget. A is not guaranteed. C is fabricated. D is incorrect — new features still require training.

**Q20. Scenario:** A manufacturing company's COO reviews the first ninety days of AI tool deployment. The team spent \$25,000 on tools and trials. ChatGPT Enterprise delivered clear ROI in quality documentation. Gong improved sales call analysis. But a specialised supply-chain AI trial showed no measurable impact after eight weeks. What is the correct ninety-day decision regarding the supply-chain AI tool?

- A) Continue the trial for another ninety days — AI takes time to show results

- B) Cancel the supply-chain tool subscription and redirect that budget to scaling the tools that demonstrated ROI
- C) Replace it with a different supply-chain AI vendor immediately
- D) Build a custom supply-chain AI tool since commercial options did not work

**Answer: B** **Explanation:** B is correct because the ninety-day strategy requires keep-or-kill decisions based on data. If a tool shows no measurable impact after eight weeks of intensive use, the rational decision is to cancel and redirect resources to proven tools. A extends an unproductive investment. C assumes the problem is the vendor rather than the use case fit. D leaps to the most expensive option after the cheapest one failed — the use case itself may not be AI-ready.